

Technical Document #169: Data Engineering - Comprehensive Overview

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Data governance establishes policies and procedures for managing data assets. It covers data privacy, security, and compliance.

Stream processing handles continuous data streams in real-time. Apache Kafka and Apache Flink are popular stream processing frameworks.

Partitioning divides data into smaller, more manageable pieces. It improves query performance by reducing the amount of data scanned.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Key Takeaways:

- Schema evolution handles changes in data structure over time.
- Data quality ensures data is accurate, complete, and consistent.
- Data pipelines automate the movement and transformation of data between systems.